

VG SERIES 3 EPISODE 2

Title: "The Mathematics of the Phase Transition"

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ABSTRACT

This episode develops the mathematical formalism of the phase transition that gives birth to a daughter Network from a black hole. Starting from the complete Hamiltonian of the Primary Network, we derive the conditions under which coupling density reaches a critical threshold, triggering a reconfiguration of spins and couplings—similar to the unpacking of the Big Bang, but occurring locally. We introduce the concept of **critical coupling density** (ρ_c), the order parameter (Φ) that distinguishes the two phases, and the phase transition can lead to three distinct outcomes, depending on the local parameters (density, rotation, spin alignment): **emergence of a new metric** in the daughter Network. The episode concludes with a set of preliminary testable predictions, including signatures in gravitational waves and the cosmic microwave background.

Keywords: phase transition, critical coupling density, order parameter, daughter Network, imprinted rotation, mathematical formalism, VG Series 3.

1. INTRODUCTION – WHY WE NEED MATHEMATICS

In Episode 1, we described the phase transition conceptually. In Episode 2, we give it a mathematical body. This is not for the sake of complexity—it is to provide:

- A **precise language** for what happens inside a black hole.
- A **bridge** to standard physics.
- A **foundation** for testable predictions.

2. THE COMPLETE HAMILTONIAN – FROM THE ONE-INCH EQUATION TO THE SPIN EXTENSION

We begin with the **one-inch equation**—the fundamental Hamiltonian of the Primary Network, first introduced in RTS16 and used throughout VG Series 2:

$$H = \sum_i \left(\frac{p_i^2}{2m} + \frac{1}{2} m \omega_i^2 q_i^2 \right) + \sum_{i < j} K_{ij} q_i q_j$$

This equation describes the individual vibrations of the nodes and the couplings between them. From it, all laws of physics emerge—including gravity, matter, and time.

In VG Series 3, we extend this equation to include spin explicitly, as a filter of permissivity. The complete Hamiltonian becomes:

$$H = \sum_i \left(\frac{p_i^2}{2m} + \frac{1}{2} m \omega_i^2 q_i^2 \right) + \sum_{i < j} K_{ij} q_i q_j + \sum_{i < j} \lambda_{ij} (\hat{S}_i \cdot \hat{S}_j) q_i q_j + \sum_i \gamma_{ij} (\hat{S}_i \times \hat{S}_j) \cdot q_i p_j$$

Where the new terms represent:

· $\lambda_{ij} (\hat{S}_i \cdot \hat{S}_j) q_i q_j$ – spin-dependent coupling, which modulates the interaction based on spin alignment.

· $\gamma_{ij} (\hat{S}_i \times \hat{S}_j) \cdot q_i p_j$ – spin-orbit coupling, which introduces rotation and chirality into the network dynamics.

Note: The one-inch equation remains the core form of the Hamiltonian. The extended version is a refinement for regimes where spin plays a critical role—such as the phase transition inside a black hole. For most cosmological and phenomenological analyses, the one-inch form remains sufficient.

3. CRITICAL COUPLING DENSITY (ρ_c)

We define the average coupling density of the Network:

$$\rho = \frac{1}{N} \sum_{i < j} |K_{ij}|$$

where **N** is the number of nodes. As matter falls into the black hole, **p** increases. When it reaches a critical value, ρ_c the Network becomes unstable.

The critical value ρ_c is determined by the maximum coupling strength that the Network can sustain before reconfiguration:

$$\rho_c = \rho_0 \cdot (1 + \alpha \cdot \langle \hat{S} \rangle^2)$$

where:

- ρ_0 = the base coupling density of the parent Network.
- α = a dimensionless constant.
- $\langle \hat{S} \rangle$ = the average spin alignment in the region (a scalar value, not an operator).

Note on notation:

The hat over $\hat{S}$ indicates that it is a quantum operator. However, $\langle \hat{S} \rangle$ denotes its expectation value—a numerical scalar. Therefore, the square is applied to the scalar $\langle \hat{S} \rangle$, not to the operator itself. This distinguishes $\langle \hat{S} \rangle^2$ (square of the mean spin) from $\langle \hat{S}^2 \rangle$ (mean of the squared spin), which would be a different physical quantity.

This equation shows that spin alignment lowers or raises the threshold—spins that are more aligned require more density to trigger the transition.

4. THE ORDER PARAMETER (Φ)

To describe the phase transition, we introduce an **order parameter** (Φ) , which measures how far the Network is from its original configuration:

- $\Phi = 0 \rightarrow$ the Network is in the normal state (no transition).
- $\Phi = 1 \rightarrow$ the Network has fully transitioned to the daughter state.

The evolution of Φ is governed by a **Ginzburg-Landau type equation**:

$$\frac{d\Phi}{dt} = -\Gamma \cdot \frac{\partial V_{\text{eff}}}{\partial \Phi}$$

where:

- Γ is a relaxation constant.
- $V_{\text{eff}}(\Phi)$ is an effective potential with two minima: one at $\Phi = 0$ (parent state) and one at $\Phi = 1$ (daughter state).

The **effective potential** is:

$$V_{\text{eff}}(\Phi) = a\Phi^2 + b\Phi^4 + c\Phi^6 + \dots$$

where a changes sign when ρ exceeds ρ_c . This is the classical signature of a phase transition.

5. THE EMERGENCE OF A NEW METRIC

When Φ shifts from 0 to 1, the coupling configuration changes. This is encoded in a **new metric** $g_{\mu\nu}$ that emerges in the daughter Network. The metric of the daughter Network is related to the parent metric by a **conformal transformation**:

$$g_{\mu\nu}^{(\text{daughter})} = \Omega^2 \cdot g_{\mu\nu}^{(\text{parent})}$$

where Ω is a scale factor determined by Φ . This transformation generates the new geometry—the new space, time, and fundamental constants.

6. IMPRINTED ROTATION – THE TORSION FIELD

If the parent black hole has angular momentum, the transition does not occur symmetrically. The order parameter Φ acquires a **phase**, which encodes the rotation:

$$\Phi \rightarrow \Phi \cdot e^{i\theta}$$

where θ is the rotation angle. This phase becomes the **torsion field** of the daughter Network—a primordial chirality that will influence the formation of galaxies, the direction of spiral arms, and the net rotation of the universe.

7. THE THREE DESTINIES – MATHEMATICAL INTERPRETATION

The phase transition can lead to three distinct outcomes, depending on the local parameters (density, rotation, spin alignment):

Destiny	Mathematical condition	Resulting regime
Pure absorption	$\rho < \rho_c$ (no transition)	Slow evaporation, may leave a shell
Birth of a bud	$\rho \geq \rho_c, \Phi \rightarrow 1, \theta \approx 0$	Daughter Network (new universe)
White hole transformation	$\rho \geq \rho_c, \Phi \rightarrow 1, \theta \neq 0$, spin inversion	Regime reversal: absorption → emission, may leave a shell after emission ceases

The phase θ encodes the rotation, and when it is non-zero, the transition preserves this rotation as a torsion field—which can sustain a white hole regime.

8. CONSERVATION OF INFORMATION – THE SUBQUANTUM WALL

During the transition, information from the parent Network is not lost. It is stored in the **subquantum wall** and transferred to the daughter Network through a **tunneling process**, represented by:

$$P_{\text{transfer}} = \exp \left(-\frac{S_{\text{action}}}{\hbar} \right)$$

where S_{action} is the action of the transition. This ensures that the daughter Network inherits the informational patterns of the parent—explaining why universes are not random, but carry memory.

9. PRELIMINARY TESTABLE PREDICTIONS

1. **Gravitational waves** – The phase transition should generate a burst of gravitational waves with a characteristic frequency spectrum, potentially detectable by future space-based observatories.
2. **CMB anomalies** – The imprinted rotation should leave a signature in the polarization of the cosmic microwave background—a net chirality or alignment of patterns.
3. **Dark matter distribution** – The fractal nature of the Network should be reflected in the large-scale structure of dark matter, with self-similar clustering patterns.
4. **Primordial black holes** – Some primordial black holes may not be relics of the early universe, but seeds of daughter Networks—micro-universes hidden within our own.
5. **Transient events** – The birth of a bud or the transition to a white hole could be observed as gamma-ray bursts or other energetic flares.

10. CONCLUSION – A MATHEMATICAL BIRTH

We have provided a preliminary mathematical foundation for the birth of a universe from a black hole:

- **Critical coupling density** (ρ_c) defines the threshold.
- **Order parameter** (Φ) tracks the transition.
- **Effective potential** (V_{eff}) determines the phase change.
- **New metric** ($g_{\mu\nu}$) emerges from the reconfiguration.
- **Imprinted rotation** becomes the torsion field of the daughter Network.
- **Information** is preserved through the subquantum wall.

These equations are a first step. They must be refined, tested, and connected to

observations. But they exist now.

11. WHAT COMES NEXT

Episode 4 of VG Series 3 will explore the **observational consequences** in greater detail: how to detect imprinted rotation in the CMB, how to model dark matter fractality, and how to search for gravitational wave signatures of the phase transition.

WHAT'S NEW IN SCIENCE – VG SERIES 3, EPISODE 2

"The Mathematics of the Phase Transition"

This episode provides the first mathematical formulation of the phase transition that gives birth to a daughter Network from a black hole. Key innovations include:

1. Critical coupling density (ρ_c) – A physical threshold defined in terms of average coupling strength and spin alignment, providing a measurable quantity that replaces the concept of singularity.
2. Order parameter (Φ) – A Ginzburg-Landau type formalism that tracks the transition from parent to daughter state, connecting the birth of a universe to established theories of phase transitions.
3. Three destinies – Pure absorption (which may leave a shell after evaporation), birth of a bud (daughter Network), or white hole transformation (which may leave a shell after emission ceases). Each destiny is determined by local parameters (density, rotation, spin alignment).
4. Imprinted rotation as torsion field – The angular momentum of the parent black hole is encoded in a phase of the order parameter, becoming the primordial torsion field of the daughter Network or sustaining a white hole regime.
5. Information preservation via tunneling – A formal expression for the transfer of information through the subquantum wall, providing a solution to the black hole information paradox within the VG framework.
6. Preliminary testable predictions – Gravitational wave signatures, CMB polarization anomalies, self-similar dark matter distribution, and transient astrophysical events as birth signals.
7. Extension of the one-inch equation – The episode introduces an extended form of the Hamiltonian that includes spin explicitly, while clearly stating that the core one-inch equation remains the fundamental form.

In one sentence: VG Series 3, Episode 2 transforms the birth of a universe from a

conceptual vision into a mathematical formalism, offering testable predictions and a bridge to standard physics